

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 10

Assume a Cross-Account Role and Create a Centralized Logging S3 Bucket

AWS STS | IAM Roles | Amazon S3 | Centralized Logging

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
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STATUS	Complete

Executive Summary

Assumed the **OrganizationAccountAccessRole** in the **SecurityAudit** AWS member account and created a new Amazon S3 bucket while operating under the temporary cross-account role session. This project established the first storage component for the portfolio's future centralized organization-wide CloudTrail logging architecture.

Business Problem

A multi-account environment needs a secure way to administer member accounts without creating separate long-term IAM users in each account. It also needs an isolated destination for centralized audit logs so security telemetry can be stored outside the management account. Cross-account role assumption and a dedicated S3 bucket in the SecurityAudit account provide the foundation for both requirements.

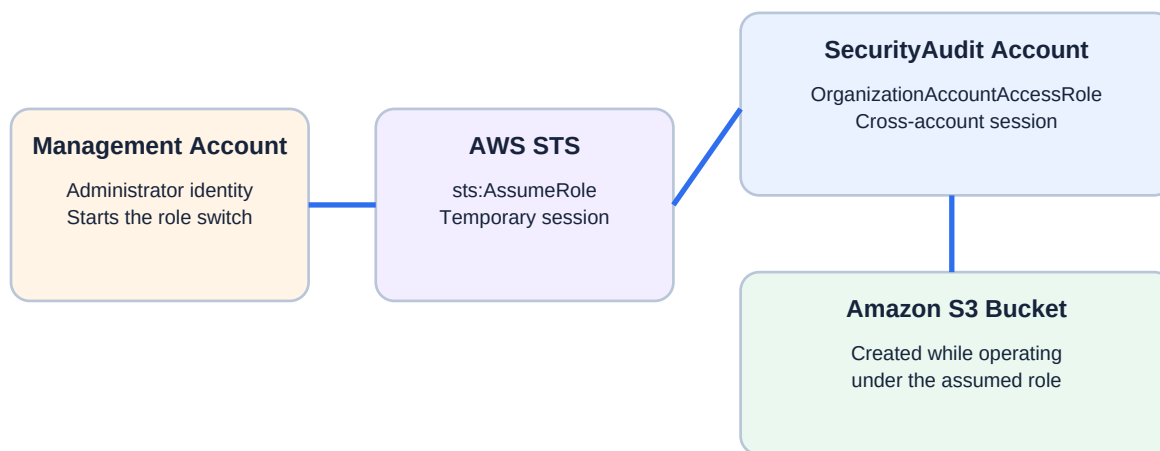
Objectives

- Switch from the management account into the SecurityAudit member account.
- Assume the OrganizationAccountAccessRole.
- Operate with temporary cross-account session credentials.
- Validate the active account and role context in the AWS console.
- Create a new Amazon S3 bucket while the SecurityAudit role session is active.
- Prepare a dedicated storage destination for future centralized CloudTrail logs.

Environment

Source Context	AWS Organizations management account
Destination Account	SecurityAudit member account
Assumed Role	OrganizationAccountAccessRole
Access Mechanism	AWS console Switch role / AWS STS AssumeRole
Credential Model	Temporary role session credentials
Resource Created	Amazon S3 general-purpose bucket
Resource Ownership	Created in SecurityAudit account context
Bucket Region	Not independently visible in the provided completion screenshot; lab guide target is us-west-2
Intended Purpose	Foundation for centralized organization-wide CloudTrail logging

Architecture



The role session changes the effective account context; resources created during the session belong to SecurityAudit.

The bucket is the foundation for later centralized organization-wide CloudTrail logging.

Figure 1 - Cross-account role assumption and centralized logging storage foundation.

Implementation Summary

1. Started from the AWS Organizations management account.
2. Used the AWS console Switch role workflow.
3. Targeted the SecurityAudit member account.
4. Assumed the OrganizationAccountAccessRole.
5. Validated that the console showed SecurityAudit as the account context and OrganizationAccountAccessRole as the active role.
6. Opened Amazon S3 while the assumed role session was active.
7. Created a new S3 bucket under the SecurityAudit account context.
8. Validated successful bucket creation and an empty object inventory.

KEY CONCEPT

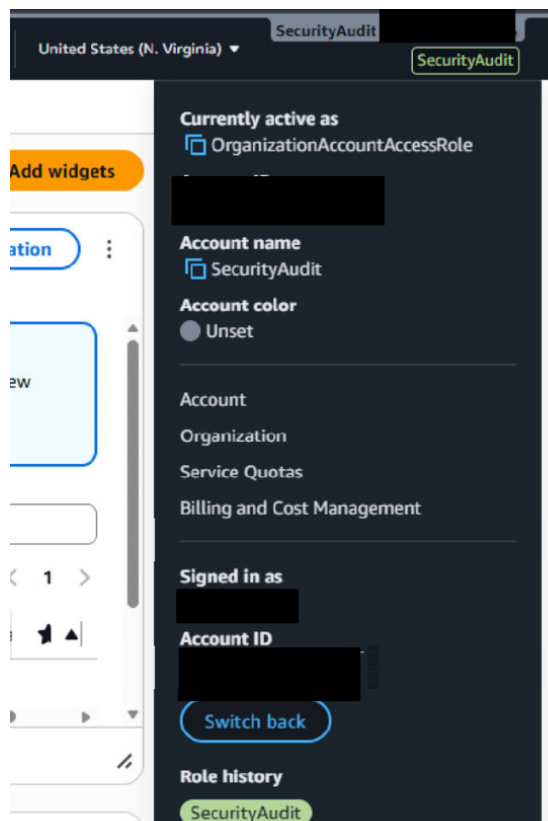
Assuming a role does not permanently change the original identity. AWS STS creates a temporary session with the destination role's permissions, and actions performed during that session occur in the destination account context.

Access Model

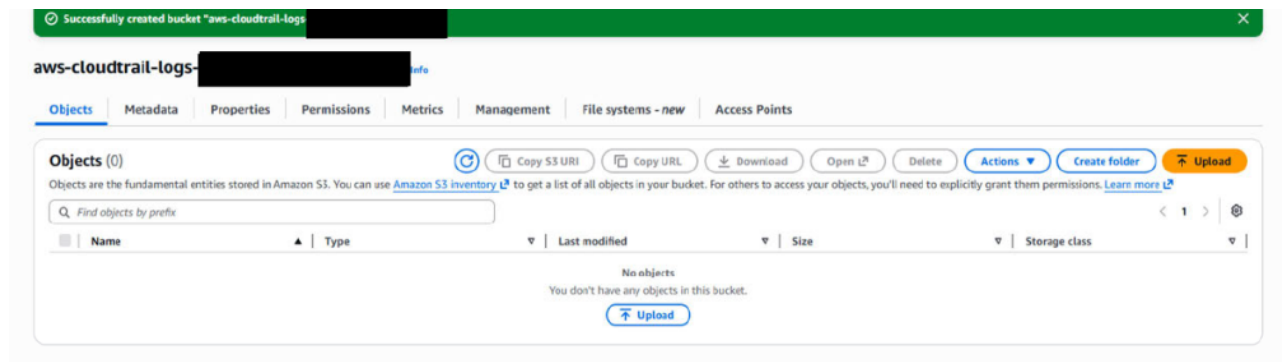
Caller Permission	The source identity must be allowed to call sts:AssumeRole
Destination Trust	The destination role must trust the source account or principal
Temporary Credentials	AWS STS issues session credentials after successful role assumption
Effective Permissions	The session uses the permissions attached to OrganizationAccountAccessRole
Resource Context	The S3 bucket created during the session belongs to SecurityAudit

Evidence

Figure 2 - OrganizationAccountAccessRole Active in SecurityAudit



The redacted AWS console identity menu shows SecurityAudit as the active account context and OrganizationAccountAccessRole as the currently active role. Sensitive account and sign-in identifiers were redacted before publication.

Figure 3 - S3 Bucket Successfully Created Under the Assumed Role

The redacted Amazon S3 console shows the successful bucket creation banner and the new empty bucket. The bucket name was partially redacted before publication.

Security Analysis

Cross-account role assumption avoids creating separate long-term IAM users in the SecurityAudit account. Access is provided through a time-limited role session instead.

The OrganizationAccountAccessRole establishes an administrative bootstrap path into the member account. Because it is highly privileged, the role should be tightly controlled and eventually complemented with narrower operational roles.

Creating the logging bucket in the SecurityAudit account separates the future audit log destination from the management account. This improves isolation and supports a more defensible centralized logging design.

Role assumption is auditable. The sts:AssumeRole event and subsequent S3 actions can later be identified in CloudTrail, which creates a useful chain of administrative accountability.

Lessons Learned

An IAM role combines permissions with a trust relationship. The caller needs permission to assume the role, and the destination role must trust the caller.

AWS STS provides temporary credentials for the assumed role session, which is safer than distributing permanent access keys across accounts.

The active account context matters. Creating a resource while operating under a role in SecurityAudit places that resource in the SecurityAudit account rather than the management account.

Centralized logging is built in stages: first create the dedicated security account and storage destination, then configure organization-wide log delivery and access controls.

Production Improvements

- Use IAM Identity Center or purpose-built cross-account roles for routine administration instead of relying broadly on OrganizationAccountAccessRole.
- Apply least-privilege permissions and short session durations to operational roles.

- Enable S3 Block Public Access and validate ownership controls, encryption, versioning, and lifecycle requirements for the centralized log bucket.
- Add a restrictive bucket policy that permits only the required CloudTrail delivery and security administration paths.
- Enable monitoring for sts:AssumeRole, S3 policy changes, bucket deletion attempts, and unexpected access to centralized logs.
- Complete the architecture by configuring an organization-wide CloudTrail to deliver logs to the SecurityAudit bucket.

Skills Demonstrated

AWS STS	Temporary credentials	Amazon S3
Role assumption	Account isolation	AWS Organizations
OrganizationAccountAccessRole	Centralized logging design	Multi-account operations

Resume Bullet

RESUME

Performed cross-account AWS administration by assuming OrganizationAccountAccessRole into a SecurityAudit member account through AWS STS and creating a dedicated S3 bucket to establish the storage foundation for centralized organization-wide CloudTrail logging.

STAR Interview Story

Situation: The multi-account AWS environment needed a secure way to administer the SecurityAudit account and prepare isolated storage for centralized audit logs.

Task: Access the member account without creating a local IAM user and create the first storage component for centralized logging.

Action: I switched roles from the management account, assumed OrganizationAccountAccessRole in SecurityAudit, validated the active role session, and created a new S3 bucket while operating under that assumed role.

Result: The environment now has a working cross-account administrative path and a dedicated SecurityAudit-owned S3 bucket ready for future organization-wide CloudTrail log delivery.

Version History

v1.0.0	July 10, 2026	Initial documented implementation. Assumed OrganizationAccountAccessRole in the SecurityAudit member account, validated the temporary cross-account session, created a new S3 bucket under the assumed role context, and embedded all provided project evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Assume the Role! (Centralized Logging, Part 1).
- User-provided redacted SecurityAudit assumed-role screenshot.
- User-provided redacted S3 bucket creation screenshot.
- Project 08 - AWS Organizations Security OU and SecurityAudit Account.
- Project 09 - Organization-Wide Service Control Policies.